DAY 1 · LOGISTICS

Is Olist's delivery network really collapsing?

> *"Review scores are falling, and Support says delay complaints are at an all-time high. Commercial is blaming the carriers. I need numbers I can defend at next Monday's board meeting."*
>
> **Marcos Silva**, Head of Logistics

40,270
delivered orders analysed

27
Brazilian states

6 months
January to June 2018

Prepared by Shatayu Roy · MySQL analysis of the public Olist dataset, January to June 2018 · Simulated case

THE BRIEF

What needed to be checked

Marcos's claims

- Are late deliveries getting worse?
- Are reviews really falling, and why?
- Is it the carriers' fault?

My own checks

- Is the national number hiding weak regions?
- Could cancelled orders explain the bad reviews?
- Did delivery really improve, or did Olist just promise longer delivery times?

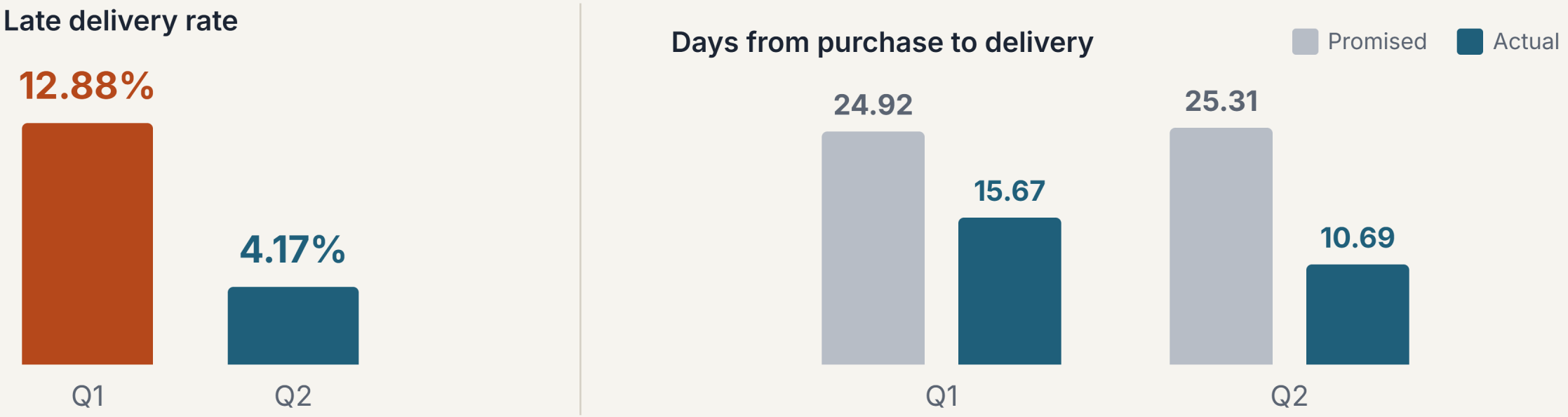
THE ANSWER

The crisis was real in Q1, and it is already recovering

1 · DELAYS	Late deliveries fell from 12.88% to 4.17%, and delivery got about 5 days faster.
2 · REVIEWS	Complaints rose and fell with late deliveries; a late order scores about 2 stars lower.
3 · CAUSE	Late orders lose their time with the carrier: 30.80 days in transit, versus 9.50 on time.
4 · STILL AT RISK	Bahia, Pará and Ceará remain above 12% late.

1 · DELAYS

Late deliveries fell from 12.88% to 4.17%, and delivery really got faster

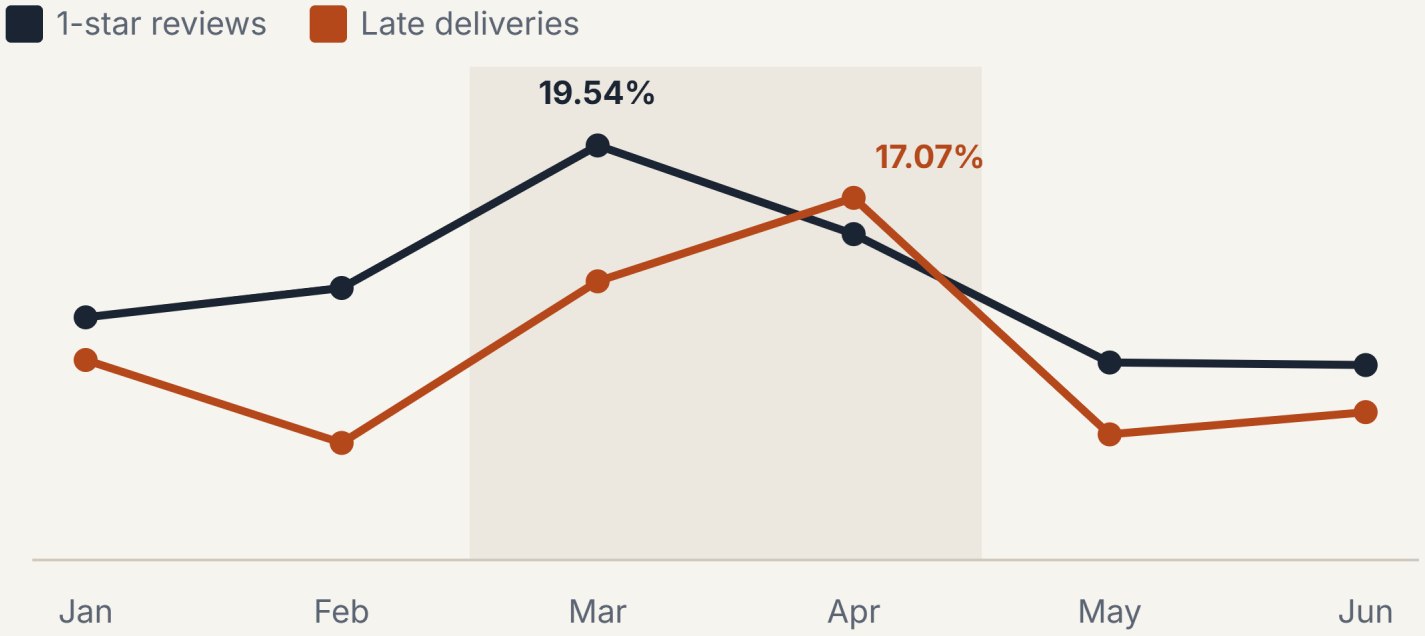


The promise stayed around 25 days while actual delivery got about 5 days faster. The drop is real.

Delivered orders grouped by purchase date · Olist delivery audit · Day 1

2 · REVIEWS

Complaints rose and fell together with late deliveries



Both peaked in March and April.

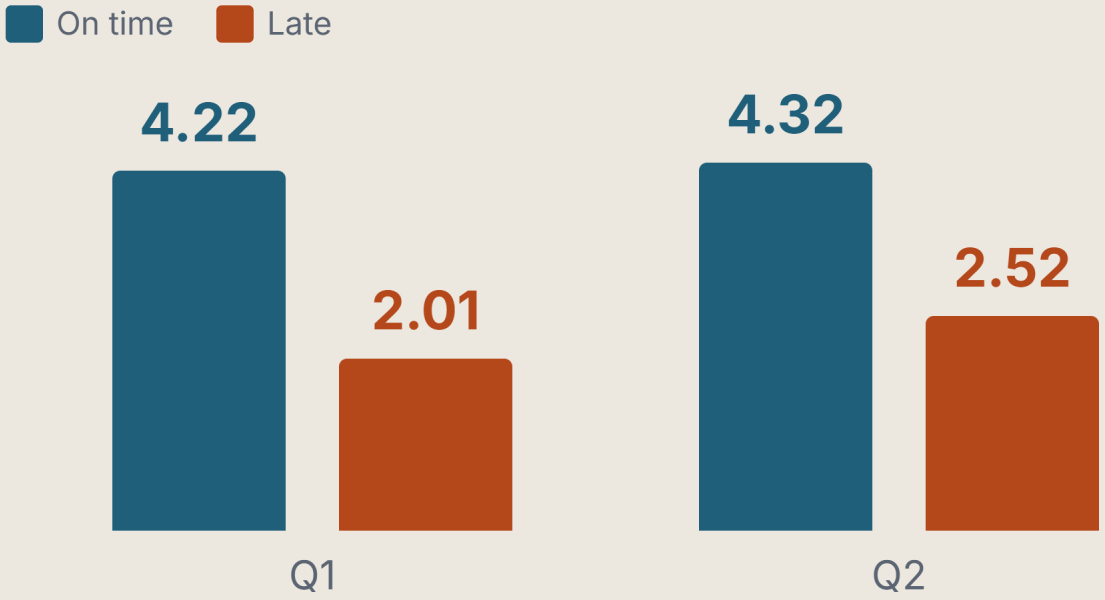
By May, late deliveries were back to 5.92% and one-star reviews to 9.31%.

The dashboard alarm came from a spike that was already ending.

Late deliveries by delivery month; 1-star reviews by review month; shaded: March and April

2 · REVIEWS

A late delivery costs about 2 stars



−2.21

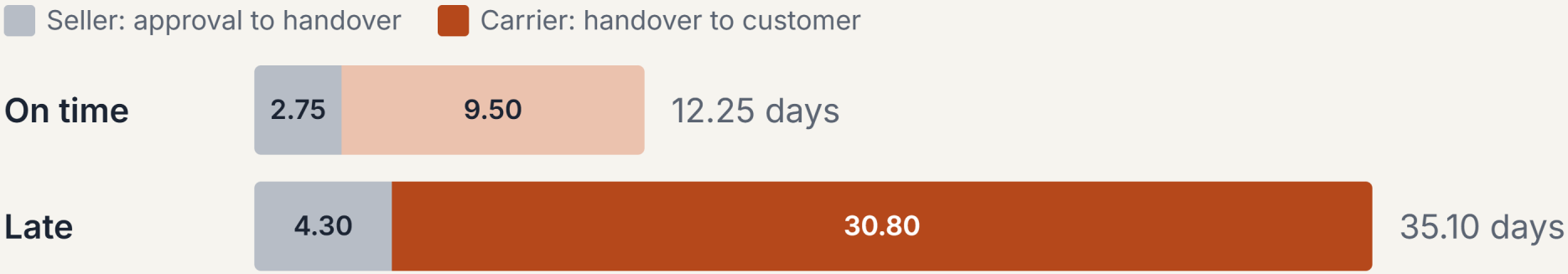
stars on average when a Q1 order arrived late (−1.80 in Q2), on a five-star scale.

Punctuality drives customer satisfaction directly.

Average review score, delivered orders with a review · Olist delivery audit · Day 1

3 · CAUSE

Late orders lose their time with the carrier, not the seller



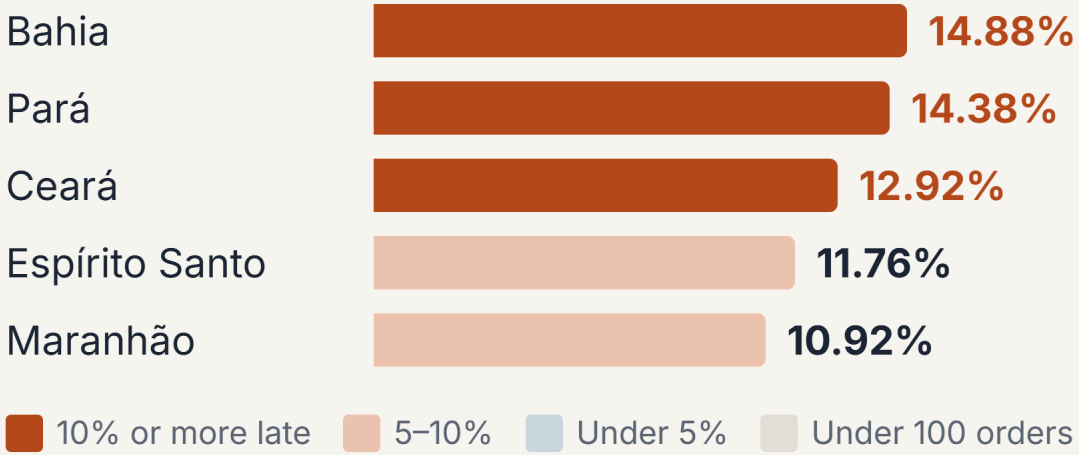
+21.30 days
more with the carrier for late orders

+1.55 days
more with the seller for late orders

Distance may explain part of this. Separating carrier quality from distance is the next check.

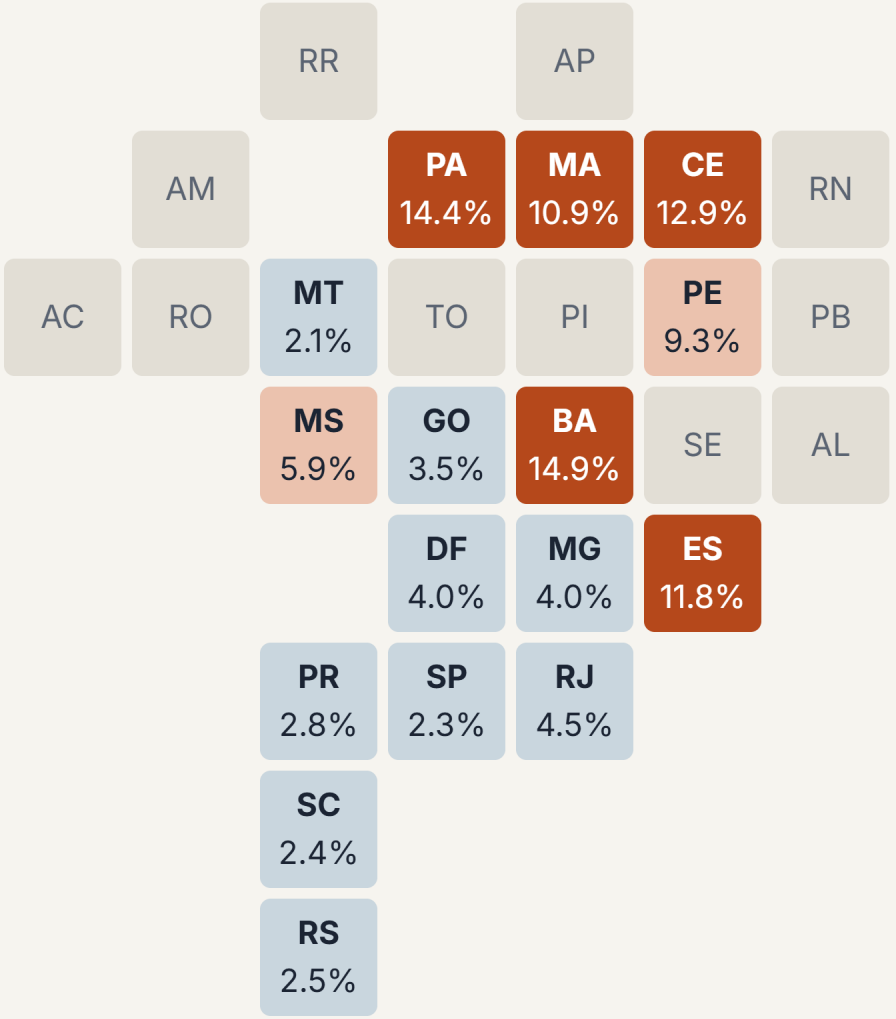
4 · STILL AT RISK

Most states recovered; Bahia, Pará and Ceará still lag



Rio, the biggest Q1 problem at 26.95% late, recovered to 4.45%.

Map: each square is one Brazilian state, placed roughly where it sits on the map.



Q2 late delivery rate by customer state, orders bought April to June 2018 · Olist delivery audit · Day 1

WHAT TO DO

Recommendations

1

Don’t overhaul all carrier contracts

Delivery is already about 5 days faster, and the late rate fell to 4.17%.

From findings 1 and 3

2

Run a carrier review in Bahia, Pará and Ceará

They are still above 12% late in Q2.

From finding 4

3

Track the monthly late rate next to review scores

Reviews show a delivery problem weeks after it happens.

From finding 2

Success measure: the three states’ late rate moving toward the national 4.17%.

WHY IT MATTERS

What the company gains from a targeted fix

~106

Fewer late orders a quarter

If Bahia, Pará and Ceará fell to the national 4.17%, their late orders would drop from about 149 to about 43.

~1.8

Stars better per order fixed

In Q2, a late order scored 2.52 stars and an on-time order 4.32.

5%

Of orders in the review

The three states held 1,038 of 19,546 Q2 orders. A focused review, not a full carrier overhaul.

Not measured here: money value. Counts are reviewed orders, as in Step 4, slightly below all orders.

NOTES

Data notes and limitations

Definitions

- Period: orders bought January to June 2018. Slide 5 groups by delivery month.
- Late: delivered after the promised date, comparing dates only.
- Orders with several reviews: scores averaged into one.

What was left out

- Orders without a review (159 in Q1, 97 in Q2), from score and state analysis.
- 321 orders with impossible dates, from timing analysis.
- States with under 100 orders, from conclusions.
- Not covered: carrier names, distance, earlier years.

BACKUP

Checks that confirmed the conclusion

−0.06

Cancelled and undelivered orders lower the overall score by only 0.06 stars in Q1, and 0.04 in Q2.

4.10 vs 3.92

Grouped by review date instead of purchase date, Q2 is still better than Q1.

196

Q2 orders still in transit, versus 264 in Q1, so slow undelivered orders don’t hide a Q2 problem.

NEXT · DAY 2 · VENDOR OPERATIONS

Should we really penalize 3,000 sellers?

Day 1 found that late orders lose about 20 extra days with the carrier, but only about 2 with the seller. The next morning, leadership wants a blanket penalty on every seller anyway.

> *"Before we trigger a seller revolt, who is actually breaking our promises? I need defensible names on my desk today."*
>
> **Camila Duarte**, Head of Vendor Operations

SQL queries, results and this deck: **github.com/shatayuroy77-netizen/olist-logistics-margin-audit**